

kind of approach some people might take with money if they were sure they were always going to have unlimited access to money. The other approach is a more “economical” approach, where you work out exactly what structure is needed in order to do whatever you’re trying to do, partly to get a better understanding of it, and partly to leave open the possibility of doing it in a category with very little structure. This more nuanced approach to category theory has greater potential for finding increasingly diverse applications in places that are less obvious, just like an economical solution to a problem is more inclusive, as it is accessible even to people who do not have money to throw around.

I think that economical category theory is more in the spirit of category theory; perhaps I just mean in *my* preferred way of doing category theory.

The process of doing category theory

So much for the motivations behind category theory. Here is what a typical process of *doing* category theory might look like, in steps.

1. We see an interesting structure somewhere.
2. We express it categorically, that is, using only objects and morphisms in some relevant category.
3. We look for that structure in other categories.
4. We use it to prove things that will immediately be true in many different categories.
5. We investigate how it transports between categories via functors.
6. We explore how to generalize it, perhaps by relaxing some conditions.
7. We investigate what universal properties it has.
8. We ask whether it has generalizations to higher dimensions via categorification.

This often involves having a slight hunch about something, following your categorical-gut instinct, and then making it precise in the language of category theory afterwards. This is why I often proposed “things to think about” that were rather vague, because part of the discipline of category theory is making precise sense of vague ideas.